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JAVA PROJECT REPORT ON

EMPLOYEE ATTENDANCE MONITORING AND LEAVE MANAGEMENT SYSTEM

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DECLARATION

I hereby declare that the project entitled "EMPLOYEE ATTENDENCE MONITORING AND LEAVE MANAGEMENT SYSTEM**"** submitted in partial fulfillment of the requirements for the course [Java Programming**]** is a **record of my original work** carried out under the guidance of **[Trainer Name]**, Technical Trainer, Technical Department, KG College of Arts and Science.

I further declare that this project has not been submitted previously, either in part or in full, for the award of any degree, diploma, or other similar title, at this or any other institution.

I have acknowledged all sources of information used in this project wherever applicable.

April 2026

Coimbatore

POOVIKA D

CERTIFICATE

This is to certify that the project entitled " EMPLOYEE ATTENDENCE MONITORING AND LEAVE MANAGEMENT SYSTEM**"** is a **bona fide work** carried out by **[POOVIKA D] [B SC COMPUTER SCIENCE]**, [COMPUTER SCIENCE**]**, **KG College of Arts & Science**, in partial fulfillment of the requirements for the course **[JAVA PROGRAMMING]**.

This project has been completed under my supervision and guidance during the academic period **April 2026**.

Place: Comibatore

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Guide: **Head of the Department**(Technical Trainer, KGCAS) (Technical Head, KGCAS)

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ABSTRACT

The **Employee Attendance Monitoring and Leave Management System** is a software application developed to efficiently manage employee attendance and leave records in an organization. The system automates the process of tracking daily attendance, calculating total working days, and maintaining leave details. It helps in reducing manual errors, saving time, and improving accuracy in record-keeping.

This system is designed using **Java programming language** and follows an object-oriented approach. It allows users to add employee details, mark attendance, apply leave, and view reports through a simple menu-driven interface. The use of data structures like ArrayList ensures efficient storage and retrieval of employee information.

The system enhances productivity by providing quick access to employee data and ensuring transparency in attendance management. It can be further upgraded with graphical user interfaces and database integration for real-time applications in organizations.

1. INTRODUCTION

IThe **Employee Attendance Monitoring and Leave Management System** is a software application designed to manage and track employee attendance and leave records efficiently. In many organizations, attendance is still recorded manually, which can lead to errors, data loss, and time consumption. This system provides an automated solution to overcome these problems.

The main purpose of this system is to maintain accurate records of employee presence, absence, and leave details. It helps administrators easily monitor employee activities and generate reports when needed. The system is developed using the **Java programming language**, which makes it reliable, secure, and easy to use.

By using this system, organizations can reduce paperwork, improve efficiency, and ensure transparency in attendance management. It also helps in better decision-making by providing quick and accurate information about employee performance and availability.

1.1 Objective

The main objectives of the **Employee Attendance Monitoring and Leave Management System** are:

- To maintain accurate records of employee attendance
- To manage and track employee leave details efficiently
- To reduce manual work and minimize human errors
- To provide quick and easy access to employee information
- To improve overall efficiency in attendance management
- To ensure transparency in employee records
- To save time and effort in maintaining data
- To generate reports for monitoring employee performance

1.2 Overview

The **Employee Attendance Monitoring and Leave Management System** is designed to simplify the process of managing employee attendance and leave records. It is a user-friendly, menu-driven application developed using Java that helps organizations maintain accurate and organized data.

The system allows the user (admin) to add employee details, mark daily attendance, apply for leave, and view employee reports. All the data is stored in a structured format using data structures like ArrayList, making it easy to access and update information.

The system works by taking inputs from the user, processing the data through predefined methods, and displaying the output in a clear format. It ensures that attendance and leave records are updated correctly and can be retrieved at any time.

Overall, this system improves efficiency, reduces manual effort, and ensures proper management of employee information in an organization.

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1.3 Features

The **Employee Attendance Monitoring and Leave Management System** includes the following key features:

- **Employee Registration**
  Allows adding and managing employee details such as ID and name
- **Attendance Tracking**
  Records daily attendance (Present/Absent) of employees
- **Leave Management**
  Enables employees to apply for leave and updates leave records
- **Data Storage**
  Stores employee information using efficient data structures (ArrayList)
- **Report Generation**
  Displays individual and overall employee attendance reports
- **User-Friendly Interface**
  Simple menu-driven system for easy operation
- **Accuracy and Reliability**
  Reduces manual errors and ensures correct data handling
- **Time Saving**
  Automates attendance and leave processes
- **Easy Data Access**
  Quickly retrieve employee details anytime
- **Scalability**
  Can be enhanced with database and GUI features

1. PROGRAM ANALYSIS

The **Employee Attendance Monitoring and Leave Management System** is developed using the Java programming language with an object-oriented approach. The program is designed as a menu-driven application that performs various operations related to employee attendance and leave management.

The system uses a class called **Employee** to store employee details such as ID, name, total working days, present days, and leave days. Another main class controls the execution of the program and handles user interaction through different methods.

2.1 Problem Definition

In many organizations, employee attendance and leave management are handled manually using registers or spreadsheets. This manual process is time-consuming, error-prone, and difficult to maintain. It often leads to issues such as incorrect attendance records, miscalculation of leave days, and difficulty in retrieving employee information.

There is no proper system to track employee presence, absence, and leave history efficiently. Managing a large number of employees becomes complex, and generating reports requires additional effort. This lack of automation reduces productivity and may affect decision-making.

Therefore, there is a need for a computerized system that can efficiently manage employee attendance and leave records. The system should provide accurate data, easy access to information, and reduce manual workload while improving overall efficiency and reliability.

2.2 Project Overview

The **Employee Attendance Monitoring and Leave Management System** is a software project developed to automate the process of managing employee attendance and leave records in an organization. This project aims to replace the traditional manual system with a computerized solution that is more efficient, accurate, and reliable.

The system is built using the **Java programming language** and follows an object-oriented approach. It provides a simple menu-driven interface through which the administrator can perform various operations such as adding employee details, marking attendance, applying leave, and viewing reports.

This project stores employee data in a structured format using data structures like ArrayList, allowing easy access, updating, and management of records. The system ensures that attendance and leave information is maintained properly and can be retrieved whenever needed.

The project focuses on improving productivity by reducing paperwork, minimizing errors, and saving time. It is suitable for small and medium organizations and can be further enhanced by integrating a graphical user interface and database for real-time applications.

2.3 Scopes and Limitations

Scope

• Managing employee details (ID, name, etc.)
• Recording daily employee attendance
• Tracking employee presence and absence
• Managing employee leave applications
• Calculating total working, present, and leave days
• Generating employee attendance reports
• Demonstrating the practical use of Java programming concepts

1. SYSTEM DESIGN

The **Employee Attendance Monitoring and Leave Management System** is designed using a simple and structured approach. It follows object-oriented principles and is implemented as a menu-driven Java application. The system is divided into different components for efficient data handling and processing.

3.1 Modules

The main modules of the **Employee Attendance Monitoring and Leave Management System** are explained below:

1. Employee Module

- **Adds new employee details**
- **Stores employee ID and name**
- **Maintains basic employee information**

2. Attendance Module

- **Records daily attendance (Present/Absent)**
- **Updates total working days**
- **Maintains attendance history**

3. Leave Management Module

- **Allows employees to apply for leave**
- **Updates leave days**
- **Keeps track of total leave taken**

4. Report Module

- **Displays individual employee details**
- **Shows attendance summary**
- **Generates overall employee reports**

5. Admin/Control Module

- **Controls system operations**
- **Provides menu options**
- **Handles user input and navigation**

6. Data Management Module

- **Stores employee records using ArrayList**
- **Handles data retrieval and updates**
- **Ensures proper data organization**

1. METHODOLOGY

The **Employee Attendance Monitoring and Leave Management System** is developed using a structured methodology to ensure efficient design and implementation. The development process follows a step-by-step approach from requirement analysis to final testing.

The development process includes the following stages:

• Problem Analysis• System Design• Program Development• Testing and Validation• Result and Documentation

4.1. Problem Analysis

The **Employee Attendance Monitoring and Leave Management System** is developed to address the challenges faced in traditional attendance and leave tracking methods. In many organizations, these processes

are handled manually, which leads to inefficiencies and errors.

Manual systems such as registers or spreadsheets require a lot of time and effort to maintain. They are prone to human errors like incorrect entries, miscalculations of leave, and data loss. It is also difficult to manage large volumes of employee data and retrieve specific information quickly.

4.2. System Design

The **Employee Attendance Monitoring and Leave Management System** is designed using a simple and efficient structure based on Object-Oriented Programming (OOP) concepts. The system is implemented as a menu-driven Java application that allows easy interaction between the user and the system.

4.3. Program Development

The **Employee Attendance Monitoring and Leave Management System** is developed using **Java** to automate the process of tracking employee attendance and leave records. It replaces manual registers and spreadsheets, reducing errors and saving time. The system is designed using **object-oriented programming**, with an Employee class to store employee details and methods to mark attendance, apply leave, and display records.

A **menu-driven interface** allows administrators to add employees, mark attendance, apply leave, and view reports. Employee data is stored in an **ArrayList**, making it easy to add, update, and retrieve records. The program was tested to ensure correct calculations and smooth operation.

4.4. Testing and Validation

Testing ensures that the program works correctly and produces the expected results.

Different test cases were used to check the functionality of the system.

Example Test Cases:

Test Case	Expected Result
Employee Login	Successfully logged in
Mark attendance(present/absent)	Attendance update correctly
Apply For Leave	Leave requst submitted successfully
Approve/Reject leave(Admin)	Leave satatus updated correctly
View attendance report	Report displayed accurately
Enter invalid leave date	Error handle properly

This phase ensures that the program runs without errors and produces accurate results.

4.5. Result and Documentation

The **Employee Attendance Monitoring and Leave Management System** works as expected. It allows administrators to add employees, mark attendance, apply leave, and view reports easily. The system accurately tracks total days, present days, and leave days for each employee.

The menu-driven interface is simple and user-friendly, and the reports clearly display employee attendance and leave information. Using an **ArrayList** ensures easy management of records. Overall, the system is reliable, reduces manual work, saves time, and improves accuracy in attendance and leave management.

4.6. Algorithm

Algorithm for the EMPLOYEE ATTENDANCE MONITORING AND LEAVE MANAGEMENT SYSTEM:

1. **Start** the program.
2. Initialize an empty **ArrayList** to store employee records.
3. Display the main **menu**with options:
4. Add Employee
5. Mark Attendance
6. Apply Leave
7. Display Employee Details
8. Display All Records
9. Exit
10. **Input** the user’s choice.
11. **If**the choice is Add Employee:
12. Input employee ID and name.

13. Create a new Employee object and add it to the ArrayList.
14. **If**the choice is Mark Attendance:
15. Input employee ID.
16. Update present days and total working days for that employee.
17. **If**the choice is Apply Leave:
18. Input employee ID and number of leave days.
19. Update leave days for that employee.
20. **If**the choice is Display Employee Details:
21. Input employee ID.
22. Show all details of that employee.
23. **If**the choice is Display All Records:
24. Loop through the ArrayList and display details of all employees.
25. Repeat steps 4–9 until the user chooses **Exit**.
26. **Stop** the program.

4.7. Flowchart

Start

↓

Display Menu

↓

Select Option

↓

Add Employee / Mark Attendance / Apply Leave / Display Employee / Display All

↓

Perform Selected Operation

↓

Go Back to Menu or Exit

↓

End

1. SOURCE CODE

```
import java.util.*;

class Employee {
    int id;
    String name;
    int totalDays = 0;
    int presentDays = 0;
    int leaveDays = 0;

    Employee(int id, String name) {
        this.id = id;
        this.name = name;
    }

    void markAttendance(boolean isPresent) {
        totalDays++;
        if (isPresent) {
            presentDays++;
        } else {
            leaveDays++;
        }
    }
```

```
    void applyLeave(int days) {
        leaveDays += days;
        totalDays += days;
    }

    void display() {
        System.out.println("\nEmployee ID: " + id);
        System.out.println("Name: " + name);
        System.out.println("Total Days: " + totalDays);
        System.out.println("Present Days: " + presentDays);
        System.out.println("Leave Days: " + leaveDays);
    }
}

public class AttendanceSystem {

    static Scanner sc = new Scanner(System.in);
    static ArrayList<Employee> empList = new ArrayList<>();

    // Add Employee
    static void addEmployee() {
        System.out.print("Enter Employee ID: ");
        int id = sc.nextInt();
        sc.nextLine();

        System.out.print("Enter Employee Name: ");
        String name = sc.nextLine();

        empList.add(new Employee(id, name));
        System.out.println("Employee Added Successfully!");
```

```
}

// Find Employee
static Employee findEmployee(int id) {
    for (Employee e : empList) {
        if (e.id == id) {
            return e;
        }
    }
    return null;
}

// Mark Attendance
static void markAttendance() {
    System.out.print("Enter Employee ID: ");
    int id = sc.nextInt();

    Employee e = findEmployee(id);
    if (e != null) {
        System.out.print("Is Present (true/false): ");
        boolean present = sc.nextBoolean();
        e.markAttendance(present);
        System.out.println("Attendance Updated!");
    } else {
        System.out.println("Employee Not Found!");
    }
}

// Apply Leave
static void applyLeave() {
```

```
    System.out.print("Enter Employee ID: ");
    int id = sc.nextInt();

    Employee e = findEmployee(id);
    if (e != null) {
        System.out.print("Enter number of leave days: ");
        int days = sc.nextInt();
        e.applyLeave(days);
        System.out.println("Leave Applied!");
    } else {
        System.out.println("Employee Not Found!");
    }
}

// Display Employee Details
static void displayEmployee() {
    System.out.print("Enter Employee ID: ");
    int id = sc.nextInt();

    Employee e = findEmployee(id);
    if (e != null) {
        e.display();
    } else {
        System.out.println("Employee Not Found!");
    }
}

// Display All Employees
static void displayAll() {
    for (Employee e : empList) {
```

```
        e.display();
    }
}

public static void main(String[] args) {

    int choice;

    do {
        System.out.println("\n===== Employee Attendance System =====");
        System.out.println("1. Add Employee");
        System.out.println("2. Mark Attendance");
        System.out.println("3. Apply Leave");
        System.out.println("4. Display Employee");
        System.out.println("5. Display All Employees");
        System.out.println("6. Exit");

        System.out.print("Enter your choice: ");
        choice = sc.nextInt();

        switch (choice) {
            case 1:
                addEmployee();
                break;
            case 2:
                markAttendance();
                break;
            case 3:
                applyLeave();
                break;
```

```
            case 4:
                displayEmployee();
                break;
            case 5:
                displayAll();
                break;
            case 6:
                System.out.println("Thank You!");
                break;
            default:
                System.out.println("Invalid Choice!");
        }

    } while (choice != 6);
  }
}
```

1. OUTPUT SCREENS

===== Employee Attendance System =====

1. Add Employee

2. Mark Attendance

3. Apply Leave

4. Display Employee

5. Display All Employees

6. Exit

Enter your choice: 1

Enter Employee ID: 101

Enter Employee Name: Ravi

Employee Added Successfully!

===== Employee Attendance System =====

1. Add Employee

2. Mark Attendance

3. Apply Leave

4. Display Employee

5. Display All Employees

6. Exit

Enter your choice: 1

Enter Employee ID: 102

Enter Employee Name: Priya

Employee Added Successfully!

===== Employee Attendance System =====

Enter your choice: 2

Enter Employee ID: 101

Is Present (true/false): true

Attendance Updated!

===== Employee Attendance System =====

Enter your choice: 2

Enter Employee ID: 102

Is Present (true/false): false

Attendance Updated!

===== Employee Attendance System =====

Enter your choice: 3

Enter Employee ID: 102

Enter number of leave days: 2

Leave Applied!

===== Employee Attendance System =====

Enter your choice: 4

Enter Employee ID: 101

Employee ID: 101

Name: Ravi

Total Days: 1

Present Days: 1

Leave Days: 0

===== Employee Attendance System =====

Enter your choice: 4

Enter Employee ID: 102

Employee ID: 102

Name: Priya

Total Days: 3

Present Days: 0

Leave Days: 3

===== Employee Attendance System =====

Enter your choice: 5

Employee ID: 101

Name: Ravi

Total Days: 1

Present Days: 1

Leave Days: 0

Employee ID: 102

Name: Priya

Total Days: 3

Present Days: 0

Leave Days: 3

===== Employee Attendance System =====

Enter your choice: 6

1. DISCUSSION & CONCLUSION

7.1 Discussion

The **Employee Attendance and Leave Management System** in Java helps organizations track employee attendance efficiently. It allows employees to be marked as present or absent and automatically updates total working days. The system also enables employees to apply for leave, ensuring leave balances are maintained accurately. Managers can view individual employee records or the status of all employees in real-time. The menu-driven interface is simple and easy to use, even for users with minimal technical knowledge. Attendance and leave data are updated immediately, reducing errors compared to manual tracking. The program uses Java concepts such as classes, objects, methods, and ArrayLists. Although it currently stores data only in memory, it can be enhanced with file or database storage. It provides a foundation for advanced HR management features like reports and GUI integration.

7.2 Conclusion

The **Employee Attendance and Leave Management System** in Java efficiently tracks attendance and leave records. It reduces manual work and ensures accurate leave management. Employees can apply for leave, and their balances are updated automatically. Managers can view individual or overall employee status easily. The menu-driven interface is simple and user-friendly. The project demonstrates the use of Java concepts like classes, objects, methods, and ArrayLists. It can be further enhanced with persistent storage and report generation. Overall, the system is a practical tool for effective employee attendance and leave management.

1. REFERENCES

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